

Input:-

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    printf("Parent process started\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("\nChild process\n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());

        printf("Child is running...\n");
        sleep(10);

        printf("Child process terminating...\n");
    }
    else {
        // Parent process
        printf("\nParent process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        printf("Parent is waiting for child...\n");
        wait(NULL);

        printf("Child terminated.\n");
        printf("Parent process terminating...\n");
    }

    return 0;
}
```

Input-:

Output-:

```
karthik@AsUs:~$ nano week3.c
karthik@AsUs:~$ gcc week3.c -o week3
karthik@AsUs:~$ ./week3
Parent process started
Parent PID : 874
Parent PPID : 362

Parent process
Parent PID : 874
Child PID : 875
Parent is waiting for child...

Child process
Child PID : 875
Child PPID : 874
Child is running...
Child process terminating...
Child terminated.
Parent process terminating...
```